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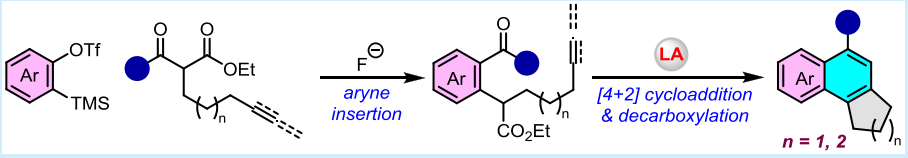
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Aromaticity-Driven Access to Cycloalkyl-Fused Naphthalenes

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ABSTRACT: We report the efficient synthesis of cycloalkyl-fused naphthalenes through the [4 + 2]-cycloaddtion/decarboxylative aromatization of alkyne-tethered aryne insertion adducts. These scaffolds were difficult to synthesize using conventional reactions. The reaction proceeds via the formation of a benzopyrylium intermediate followed by intramolecular [4 + 2] cycloaddition and a subsequent decarboxylation pathway. This method is also compatible with allene-tethered substrates to afford similar products. In addition, the one-pot synthesis of polysubstituted naphthalenes via aryne insertion/benzannulation has also been developed in good yield.

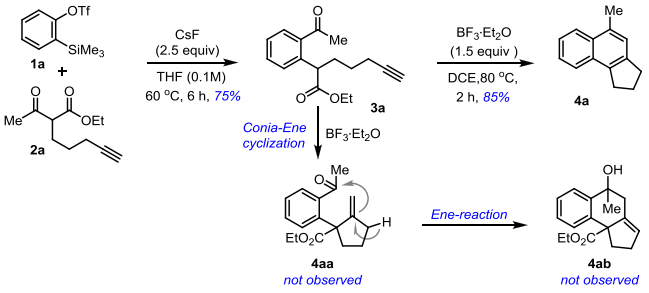
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The construction of aromatic architectures with predefined substitutions is an important area of research.[1] Polycyclic aromatic compounds have useful applications in material science.[2] Naphthalene chemistry has played a pivotal role in asymmetric catalysis.[3] The Naphtha byproducts of petroleum refinery have become the useful starting materials in this research activity. The [4 + 2] cycloaddition is most sought-after reaction to annulate the existing benzene rings.[4] These cycloaddition reactions need a standard diene/dienophile with a favorable electronic configuration. The benzannulation reactions are reported involving transition-metal catalysis,[5] Lewis acid catalysis,[6] and thermal cycloadditions.[7]

Benzannulation to construct substituted naphthalenes could provide varied substitutions (mono, di, tri, etc.) based on the reactants.[8] Our group has been expanding the scope and applications of Stoltz's cycloaddition−fragmentation process to natural product synthesis and toward medicinally relevant bioactives.[9] Stoltz's process is basically the addition of a β-keto ester to an in-situ-generated benzyne to realize an ortho-disubstituted benzene with a carbonyl group and a methylene ester taking the ortho positions to each other (acyl-phenyl acetic acid derivatives).[10] We anticipated that an appropriate substitution on the C-2 position of the β-keto ester having an alkyne group (pentyne or hexyne) would trigger a Conia-Ene process followed by a cascade of reactions to provide a functionalized tricyclic framework.

The study was initiated by the cyclization of **3a** (readily synthesized from acyl alkylation of the benzyne precursor **1a** with β-ketoester **2a**) in the presence of $BF_3{\cdot}Et_2O$ (1.5 equiv) in 1,2-dichloroethane (DCE) at 80 °C for 2 h to provide the product **4a** in 85% yield (Scheme 1). The product **4a** was confirmed by nuclear magnetic resonance (NMR) spectroscopy and high-resolution mass spectrometry (HRMS) analysis.[11] Instead of alicyclic rings (two) being stitched (**4ab**) via the

Scheme 1. Initial Results



Conia-Ene product **4aa**, we observed a clean naphthyl-fused cyclopentane **4a** via [4 + 2]-cycloaddition in decent yield. With this initial observation, we screened the annulation reaction with a catalytic amount of various Lewis acids including $BF_3{\cdot}Et_2O$ (see the Supporting Information for details) and obtained 0 to <10% of product; in most cases, the starting material was recovered. Among all verified reaction conditions, 1.5 equiv of $BF_3{\cdot}OEt_2$ in DCE at 80 °C for 2 h provided the best yield.

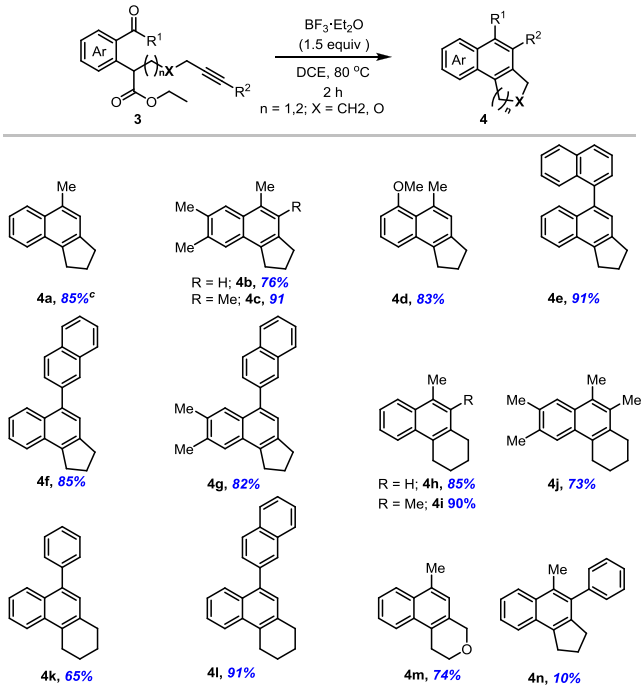
With the optimized conditions in hand, we then explored the substrate scope with various alkyne-tethered aryl ketones **3**, and the results are summarized in Scheme 2. Initially, the reaction was carried out with pentyne-tethered aryl ketones, where R^1 is an alkyl, aryl, or naphthyl, giving the corresponding

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Scheme 2. Substrate Scope for Alkynes[a,b]

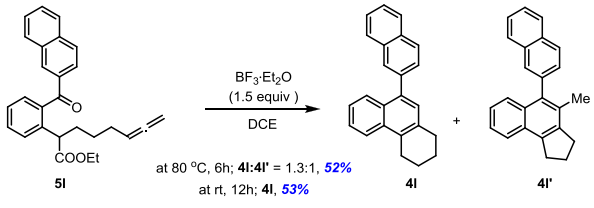


[a]Reaction conditions: **3** (0.1 mmol) and $BF_3{\cdot}Et_2O$ (0.15 mmol) in DCE (0.2 M) at 80 °C under a N_2 atmosphere. [b]Isolated yields. [c]Compound **4a** was synthesized on a 364 mg scale in 80% yield.

cyclopentane-fused naphthalenes **4a**–**4g** in good to excellent yields. It is interesting to note that the methyl-substituted alkyne was also well tolerated in the reaction (entries **4c**, **4i**, and **4j**). The [4 + 2]-cycloaddition of hexyne-tethered aryl ketones also furnished the corresponding fused cyclohexane products **4h**–**4l** in 53–91% yield. For the *O*-tethered alkyne, where R^1 was Me, the reaction proceeded well to offer the annulation product **4m** in 74% yield. When R^1 was a methyl and R^2 was a phenyl, the reaction proceeded to give the cyclized product **4n** in 10% yield. Unfortunately, further attempts to synthesize four- and seven-membered ring- fused naphthalenes were not successful.

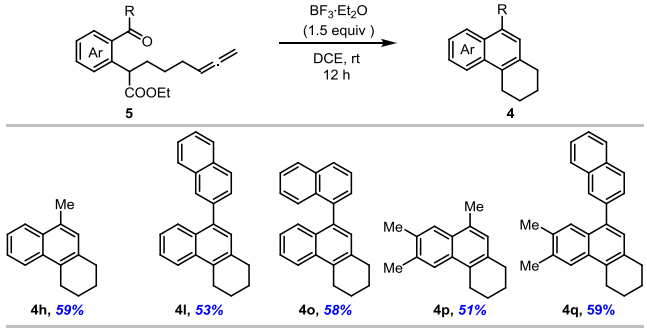
Our next task was to explore the intramolecular [4 + 2]-benzannulation with allene-tethered aryl ketones. Initially, allene **5l** was subjected to standard reaction condition to give a mixture of cyclohexane- and cyclopentane-fused products **4l** and **4l′** in a 1.3:1 ratio as an inseparable mixture in 52% overall yield (Scheme 3). The same reaction at room temperature exclusively afforded the cyclohexane-fused product **4l** in 53% yield. Steric hindrance is the determining factor behind the observed regioselectivity for the allene terminal double bond. Various allenes **5** with different aryl and alkyl substituents at the 1-phenyl ketone proved to be suitable substrates for the

Scheme 3. Reaction with Allene-Tethered Substrate



annulation reaction at room temperature to give the corresponding products (**4h**, **4l**, **4o**, **4p**, and **4q**), albeit in moderate yield (Scheme 4).

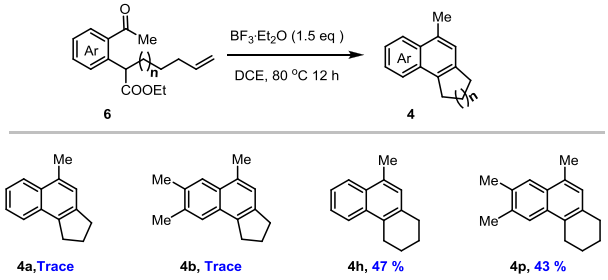
Scheme 4. Substrate Scope for Allenes[a,b]



[a]Reaction conditions: **5** (0.2 mmol) and $BF_3{\cdot}Et_2O$ (0.3 mmol) in DCE (0.2 M) at room temperature under a N_2 atmosphere. [b]Isolated yields.

Next, we explored the scope of alkene-tethered aryl ketones **6** in [4 + 2]-benzannulation under the standard reaction conditions at 80 °C (Scheme 5). To our delight, hexene-

Scheme 5. Substrate Scope for Alkenes[a,b]



[a]Reaction conditions: **6** (0.2 mmol) and $BF_3{\cdot}Et_2O$ (0.3 mmol) in DCE (0.2 M) at 80 °C under a N_2 atmosphere. [b]Isolated yields.

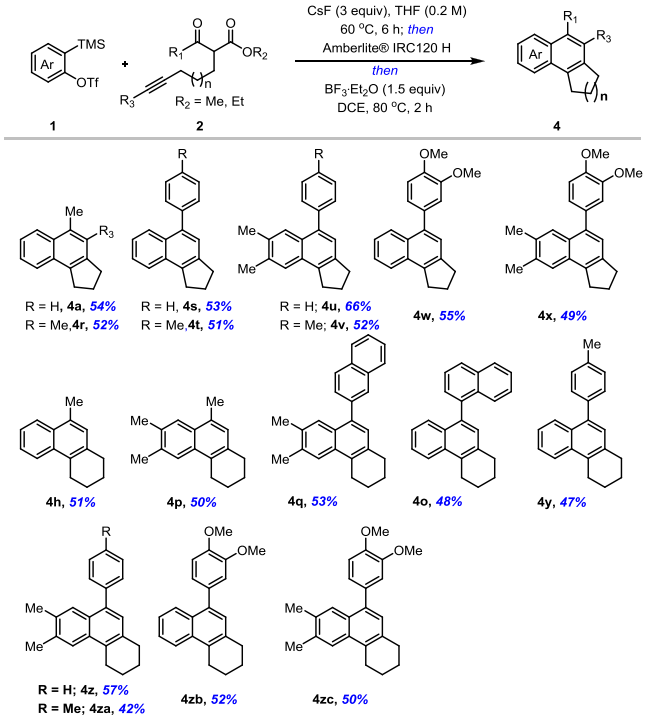
tethered substrates gave the corresponding naphthalenes **4h** and **4p** in moderate yield. However, in the case of pentene-tethered substrates, we observed only trace amount of products (entries **4a** and **4b**).

Enticed by the previously described experimental result, we attempted the one-pot aryne insertion/benzannulation reaction between the aryne precursor and the alkyne-tethered *β*-keto ester by the sequential addition of reagents (Scheme 6). Initially, triflate **1** and *β*-ketoester **2** were treated with CsF (3 equiv) in tetrahydrofuran (THF) solvent at 60 °C for 6 h. After completion of the insertion reaction (monitored by thin layer chromatography (TLC)), the excess CsF was quenched with the addition of Amberlite IR120 H until the reaction mixture became acidic; then, the reaction mixture was filtered and evaporated under reduced pressure. The crude product was dissolved in DCE; then, $BF_3{\cdot}Et_2O$ (1.5 equiv) was added, and the reaction was continued at 80 °C for 2 h. To our satisfaction, we were able to isolate the required products in moderate to good yields, as shown in Scheme 6.

On the basis of the above experimental results and previous reports, a plausible reaction pathway is proposed in Scheme 7. The reaction might be initiated by the activation of alkyne **3** or allene **5** (**A**) with Lewis acid to form the hemiketal

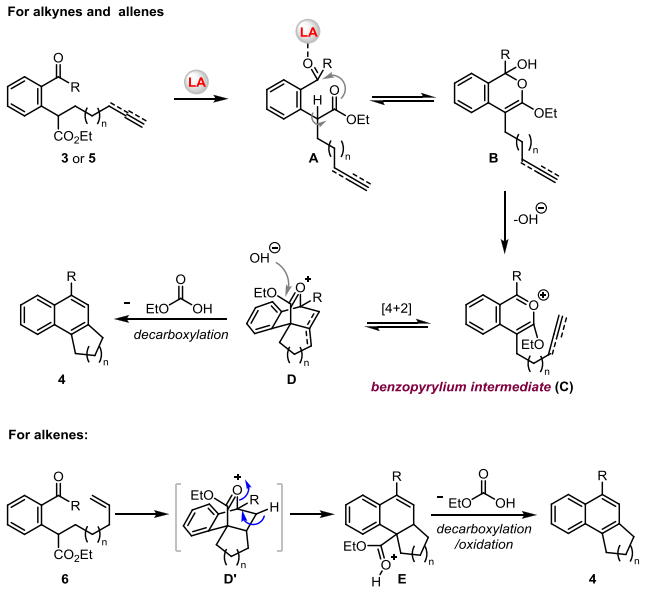
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Scheme 6. One-Pot Aryne Insertion/Benzannulation Reactions[a,b]



[a]Reaction conditions: **1** (0.12 mmol), **2** (0.1 mmol), CsF (0.9 mmol), and THF (0.2 M) at 60 °C under a N_2 atmosphere; Amberlite IR 120 H (150 mg); $BF_3{\cdot}Et_2O$ (0.45 mmol) in DCE (0.2 M); [b]Isolated yields.

Scheme 7. Plausible Reaction Pathway



intermediate **B**. The elimination of the hydroxyl group from intermediate **B** affords the highly reactive benzopyrylium intermediate **C**, which readily undergoes intramolecular [4 + 2] cycloaddition to give the bridged intermediate **D** followed by decarboxylation to produce the required naphthalene product **4**. In the case of alkene **6**, intermediate **D′** undergoes ring-opening via β-H elimination[11] to give intermediate **E** followed by acid-mediated decarboxylation/oxidation[12] to provide the desired aromatized product **4**.

In conclusion, we have developed the Lewis-acid-catalyzed intramolecular [4 + 2]-cycloaddition/aromatization of alkyne-, allene-, and alkene-tethered aryne insertion adducts to access cycloalkyl-fused naphthalenes. The reaction pathway involves the formation of a benzopyrylium intermediate followed by intramolecular [4 + 2] cycloaddition and subsequent decarboxylation. In addition, a one-pot aryne insertion/benzannulation reaction has also been developed in good yield.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c01220.

Experimental procedures, characterization details, and ^1H and ^{13}C NMR spectra of new compounds (PDF)

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Author Contributions

[§]K.S. and S.R. contributed equally to the work.

Notes

The authors declare no competing financial interest.

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